Project Executable Files

Date	4 July 2026
Team ID	SWTID-2026-8204
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA — no database used; CSV dataset + model.pkl file storage instead

5	Environment configuration file (.env.example or similar)	NA — no environment variables/secrets required
6	Deployed application URL (if hosted)	No — application currently runs locally only
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA — web application, not mobile/desktop
8	Dockerfile / Containerization config (if applicable)	No — not containerized in current version
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

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Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

Step 3: Deployment / Access Details

S.No	Known Issue / Limitation	Workaround / Status

1	Application currently runs only on Flask's development server (not production-ready)	Plan to deploy using Gunicorn + Nginx or a cloud platform (Heroku/AWS) in future iteration

Field	**Details**
Hosted / Deployed URL	
Login Credentials (Demo)	[Provide demo username and password for evaluator access]
Platform / Hosting Provider	[e.g., Render, Vercel, AWS, Heroku, Azure]
Repository Link	https://github.com/KusumaReddyYarram/opti_crop
Demo Video Link	https://drive.google.com/file/d/1VTdAAhdNiXOv6w99EHCsJ4D07h78VyMv/view

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

Pre-requisites: Python 3.x, pip, Anaconda Navigator (recommended) or virtualenv

1. Clone the repository: git clone <repository-url>
2. Navigate to the project folder: cd OptiCrop
3. Install dependencies: pip install -r requirements.txt
4. Ensure model.pkl is present in the /model folder
5. Start the application: python app.py
6. Open browser at http://127.0.0.1:5000/

Step 5: Known Issues / Limitations

2	No user authentication or saved prediction history	Out of current project scope; can be added in future versions

3	Model trained on a fixed dataset; does not autoupdate with new agricultural data	Future enhancement: enable periodic retraining pipeline with updated datasets